

UI Requirements for Intent-Aware Routing Component

This UI must clearly display **Before → After** behavior across all three layers of the intelligent routing system.

1 Telemetry Layer (Before vs After)

Telemetry = live network metrics collected from Component 1 (eBPF telemetry).

BEFORE (Healthy State)

Display baseline network conditions:

- DNS resolution time (e.g., **23 ms**)
- TCP handshake latency (e.g., **80 ms**)
- Packet retransmissions (e.g., **0**)
- Active backend route (e.g., **Service B1**)

AFTER (Degraded State)

Show updated metrics when performance drops:

- DNS delay increased (e.g., **240 ms**)
- TCP handshake latency **exceeds threshold**
(e.g., **180 ms > 100 ms**)
- Packet retransmissions increased (e.g., **0 → 7**)
- Active backend route changed
(e.g., **Service B1 → Service B2**)

Purpose

This section must make it clear:

- What the telemetry was **before** the violation.
- What changed that **triggered** the intent evaluation.

These metrics match the research plan requirement to log: **DNS delay, TCP latency, retransmissions.**

2 Intent Decision Layer (Before vs After)

This section shows how the system evaluates telemetry **against the intent policy**.

BEFORE (No Violation)

Show:

- Loaded intent policy:

```
{ "intent": "low-latency", "threshold_ms": 100, "action": "reroute" }
```

- Decision: **"Within threshold → No action"**
- Comparison details:
 - Measured latency (e.g., **80 ms**)
 - Threshold (e.g., **100 ms**)

AFTER (Violation Detected)

Show:

- Trigger metric:
(e.g., **Latency = 150 ms > 100 ms**)
- Decision: **"Threshold violated → REROUTE triggered"**
- Highlight violation (e.g., red indicator, alert banner)

Purpose

This proves:

- The system is **intent-aware**.
- The routing engine compares telemetry with policy rules.

This aligns with the **Routing Intelligence Layer** of the research plan.

3 Routing Enforcement Layer (Before vs After)

This section visualizes the **actual routing action** taken by the eBPF layer.

BEFORE (Normal Routing)

Show:

- Active route: **Service B1**
- No reroute triggered
- Kernel logs (optional):
 - **"Intercepted TCP connection"**

AFTER (Reroute Applied)

Show:

- New route: **Service B2**
- Reroute event timestamp
- Kernel/eBPF logs confirming the action
- (Optional) Improved metrics after reroute:
 - e.g., latency drops from **180 ms → 73 ms**

Purpose

This visualizes the core of the research:

- **Real-time, kernel-level service-to-service routing**
 - Based on intent evaluation + telemetry changes
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Summary for UI Developer

Your UI must contain **three sections**, each showing **Before vs After**:

A. Telemetry View

- DNS delay, TCP latency, retransmissions, active backend
- Clearly highlight changes that caused the intent violation

B. Intent Decision View

- Intent policy (thresholds + action)
- System decision: no action vs reroute triggered
- Highlight threshold violations

C. Routing Enforcement View

- Routing path before (Service B1)
 - Routing path after (Service B2)
 - Reroute timestamp
 - Kernel log confirmations
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Overall Flow Illustrated by the UI

1. **What was happening?** → Telemetry
2. **What did the system decide?** → Intent Decision
3. **What changed in the network?** → Routing Enforcement